SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	SPARK SURGE
Project / Product Name	LARIS: Quotation Tracker for Malaysian SMEs
Final Live Deployment URL	https://laris-malaysia.replit.app
Codebase Repository URL	https://github.com/lkaixian/laris
Name (Leader)	HARRISH GOVIND A/L SETHULINGAM
Email & phone (Leader)	harrishgovind09@gmail.com · [0125220471]

2. Project Description

[Maximum 250 words]

Malaysia has about 1.07 million micro, small and medium enterprises (MSMEs), which produce 39.7% of GDP and employ 48.7% of workers (DOSM). In made-to-order trades such as signage, printing and 3D printing, every job starts with a quote. Owners build it by hand from chats, supplier PDFs and remembered prices, so replies are slow, stale costs erode margin and unanswered quotes are never chased. Average business payment delay is 61 days (Experian 2025).

LARIS is a quotation tracker that works inside the chat apps owners and customers already use. A customer sends a text or a Manglish voice note on Telegram (WhatsApp is built but not yet connected to Meta). ILMU turns it into items and quantities. A rules engine, never the AI, prices the quote from the owner's cost book and applies a margin floor. The owner approves, and LARIS sends the quote (PDF, XLSX or CSV), tracks it and chases unanswered ones. The customer accepts by link, which creates a job card.

Staff can photograph a shelf or price list: Qwen2.5-VL on a RunPod GPU, cross-checked by OCR, turns it into catalog items that customers can ask about and request quotes for.

Since Phase 1 we extended the loop past acceptance: proforma and deposit, delivery order signed by tap, invoice, credit note, goods received and a revenue page. It has roles, an audit log, PDPA tools, API keys and four languages. It supports SDG 8.3: less admin, so small firms win more work and grow.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Flow: customer or staff message (Telegram live, WhatsApp client built) → FastAPI webhook on Replit → ILMU (speech to text, item extraction) or RunPod vLLM (catalog photos) + Tesseract cross-check → rules engine + PostgreSQL cost book → draft quote → owner approval in the web app → quote sent to chat, tracked, exported → customer accepts by link → job card → order documents.

- Frontend: React 19, TypeScript, Vite and Tailwind CSS 4 owner web app, plus public customer pages (accept quote, sign delivery, track), in English, Malay, Chinese and Tamil.

- Backend: Python FastAPI (uvicorn) on a Replit deployment. APScheduler runs expiry, follow-up reminders and the chat retention purge. Documents are made with reportlab (PDF, Tamil shaping via uharfbuzz) and openpyxl (XLSX).

- GenAI: ILMU ilmu-asr-v4.2 transcribes Manglish voice notes. ilmu-mini-v3.3 returns item keys and quantities as JSON. Qwen2.5-VL-7B-Instruct on vLLM (RunPod GPU) reads catalog photos, and OCR must find the same digits before a photo price is kept. The AI never sets a price: src/pricing.py rejects price-like keys, and the database enum price_src has no AI value.

- Database: PostgreSQL 16 with row-level security per business (laris_app role), append-only payments and document transitions, and an audit log. Tests run on PGlite and on a real Postgres 16 service in CI.

- Integration: API keys for a read-only recipe list and a price check priced by the same rules (/api/v1), plus a read-only /api/mcp. No Orbyte API was available, so none was called.

- Control: owner approval, custom roles (staff cannot see cost or margin), confidence checks, activity log.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

Built and deployed at laris-malaysia.replit.app. Git history for the sprint window (31 Oct 09:30 to 1 Nov 01:30): 98 non-merge commits from five contributors.

- Order chain after acceptance: proforma with a deposit (50% shown by default), deposit receipt, delivery order the customer signs by tap on a public page, invoice, balance receipt, credit notes (reduce Revenue sales and carry SST; an invoice cannot be voided under a live credit note) and goods received against supplier POs.

- Deals page (an order's money in one place, overdue rules) and Revenue page (gross and net profit by period).

- Layout engine for customer PDFs: one document language per document, Chinese and Tamil fonts.

- Pay-online license: Stripe Checkout, and a signed webhook issues and emails the key.

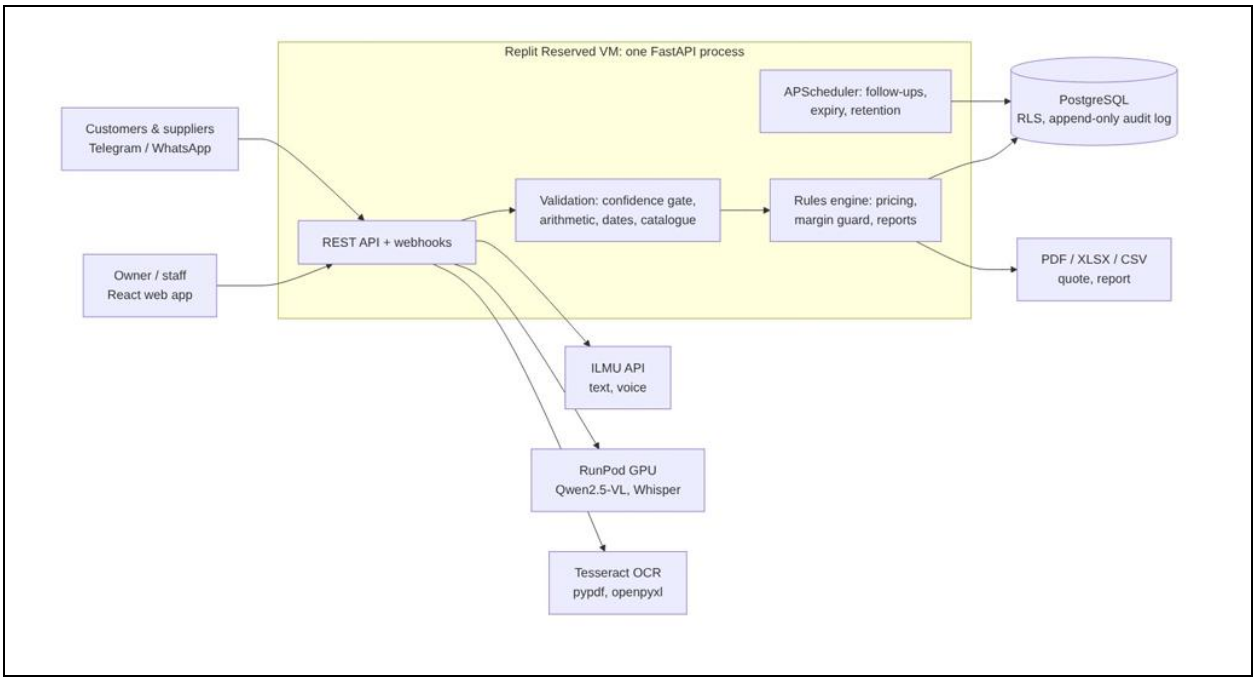
- 3D printing edition, stage 1: STL reading and a price engine under the same no-AI-price rule.

- Image-analysis meter against the Orbyte plan tiers (30, 150, 500 a month; shown, not enforced), and a written second-business scenario.

- Security: pypdf upgraded after 7 known vulnerabilities. Edge-case tests found and fixed three licence bugs.

Pivot from Phase 1: the prototype stopped when a customer accepted a quote, yet the baseline pain is late payment (61 days). We extended the loop to deposit, delivery proof, invoice and credit note, and kept every step behind the same owner approval and audit log.

Honest limits: e-signature is simulated and labelled so. Payments are records, not bank transactions. WhatsApp is not connected to Meta. Malay, Chinese and Tamil wording is machine-translated and awaits native review. PDPA tools are not lawyer-reviewed.



4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	**Your Response**
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	Phase 1 baseline: manual minutes per quote were not measured, so we claim no percentage reduction. Context baseline: 61-day average payment delay (Experian 2025). Phase 1 targets: minutes from enquiry to sent quote −50%; 90% of enquiries answered within 1 hour; zero sent quotes on a stale cost without warning.

Metric Category	Your Response
	Phase 2, measured: - Speed to quote is instrumented. Home shows the median minutes from quote created to sent (GET /api/dashboard quote_speed). There is no manual baseline to compare it with yet. The 1-hour first-reply KPI is not instrumented. - In the test fixture, a supplier price rise (acrylic sheet RM100.00 → RM140.00) flags the already-sent quote with a margin delta of −RM400.00 (test_stale_alert_shows_400_delta). - 0 prices come from AI. Three price-injection variants are rejected with 422 and no row is created. - Illustrative model (assumptions, not results): 30 quotes a month × 10 minutes saved × RM0.1453 a minute (minimum wage RM8.72 an hour) = RM43.60 a month, against a RM39 price hypothesis. Variable cost is about RM0.07 a quote (estimate).
Verification Evidence Method *How was this data verified? e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	- 1,123 pytest tests in 72 files run in GitHub Actions on every push as parallel jobs: four backend groups on PGlite, the same four on a real PostgreSQL 16 service, a checks job (pip-audit, schema, database and role checks) and a web build. The latest master run (707314e) passed. - Row-level isolation and role limits are tested, for example test_staff_never_sees_cost_or_margin. - Every step is a numbered row in docs/BUILD_LOG.md (236 rows) with its test result, and a commit in the public repo. - Limits: all data is synthetic. ILMU accuracy on real Manglish audio, OCR accuracy on real quotes and minutes saved at real shops are not yet measured (protocol in docs/eval-manglish.md).
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world*	- The target users are made-to-order micro and small firms that quote from chat.

Metric Category	Your Response
solution capable of being supported by civic or corporate partners?	- LARIS replaces retyping, remembered prices and manual chasing with a draft the owner approves. It runs on a phone and a chat app, with no ERP, in four languages. - SME support bodies or trade associations could offer licences to their members. No partner has been contacted. - Privacy tools are built (consent notice, /mydata and /forget, breach register with deadline clocks) but not lawyer-reviewed. That review comes before any real customer data.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	- A signed software licence (Ed25519) and Stripe Checkout purchase are built. - Pricing: Lite RM59 a month or RM590 a year, Standard RM189 or RM1,890, Pro RM499 or RM4,990. Yearly is 10× monthly, so two months are free. These prices are hypotheses, not validated with owners. - Variable cost is about RM0.07 a quote (estimate). The GPU pod is billed only while running (US$0.59 an hour). - It also works as an API service: the /api/v1 price check and a read-only MCP endpoint can be called by other systems. A usage meter counts quotes and image analyses against plan tiers. - 30 quotes × 10 minutes × RM0.1453 = RM43.60 a month at the wage floor. That is below the Lite price (RM59), so the benefit case rests on owners' own time being worth more than the minimum wage, and it is untested.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

Rule 1: the AI cannot price. src/pricing.py

PRICE_KEYS =
{"price","unit_price","total","amount","discount","cost","unit_cost","markup","markup_pct"
}

def _reject_price_keys(item):

bad = {k for k in item if k.lower() in PRICE_KEYS}

bad |= {k for k in item.get("attrs") or {} if k.lower() in PRICE_KEYS}

if bad: raise PriceInjection(f"price keys not allowed in input: {sorted(bad)}")

db/schema.sql

CREATE TYPE price_src AS ENUM
('cost_item_markup','owner_override','fixed_rate_card'); -- never 'ai'

Rule 2: price = cost x (1 + markup/100), checked by the margin guard

*def _pct(rev, cost): return None if rev == 0 else ((rev - cost) / rev * 100).quantize(CENT, ROUND_HALF_UP)*

def margin_guard(quote, min_margin_pct): # a line or quote below the floor is returned, never sent silently

for l in quote["lines"]:

m = _pct(l["line_total"], l["line_cost"])

if m is not None and m < floor: out.append({"line_no": l["line_no"], "margin_pct": m, "min_margin_pct": floor})

Run output (cost RM10.00, markup 80%):

- Unit price = 10.00 × 1.80 = RM18.00.

- Margin = (18.00 − 10.00) / 18.00 = 44.44%.

- margin_guard(quote, 30) returns [], so it passes.

- margin_guard(quote, 50) returns [{line_no: 1, margin_pct: 44.44, min_margin_pct: 50}, {line_no: None, ...}], so it is flagged for the owner.

Rule 3: a supplier price rise is caught on sent quotes. Test fixture: acrylic_3mm_sheet cost RM100.00 → RM140.00. GET /api/alerts/stale-costs returns margin_delta -400.0000, and the tracker sets cost_is_stale = true (tests/test_api.py).

Time value (illustrative): 30 quotes × 10 min × RM8.72 / 60 = RM43.60 a month.

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

Funding would turn a working prototype into a measured service. In order:

1. Prove the benefit. Time manual quotes against LARIS quotes at 10 to 20 owner shops across two or three trades, and publish the real minutes saved. Today we have a model, not a measurement.

2. Connect WhatsApp through Meta (template approval, business verification), because most owners and customers quote there. Telegram stays the fallback.

3. Compliance before real data. A Malaysian lawyer reviews the PDPA tools (brief ready, 11 questions), and we sign data processing terms with each vendor.

4. Scale the platform. Move catalog reading from a stopped-when-idle GPU pod to serverless with a warm option. Run the app as the row-level-security role on managed PostgreSQL. Measure ILMU and OCR accuracy on real Manglish audio and real quotes.

5. Distribution. A new trade is one YAML file, so partners such as SME support bodies, trade associations or Rotary clubs can onboard members by trade. The /api/v1 and MCP endpoints let platforms such as Orbyte offer LARIS to their customers.

6. Business model. Three tiers, Lite (RM59 a month, RM590 a year), Standard (RM189, RM1,890) and Pro (RM499, RM4,990), plus optional per-quote add-ons. Prices are hypotheses to test in the pilot. As a scenario only, 100 Lite businesses on monthly billing would be RM70,800 a year. Variable cost is about RM0.07 a quote.

After that: supplier price feeds, then Singapore and Indonesia.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- [X] The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- [X] The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- [X] The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- [X] The final pitch deck has been exported and saved as a flattened PDF file.
- [X] All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No [x] No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

[x] Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
[ALL EXISTING PHASE 1 DECLARATION THIRD-PARTY CODE/DATASET/MODEL]		
uharfbuzz 0.57.0	**code**	**Apache-2.0**
pypdf 6.21.4 (was 6.21.2)	**code**	**BSD-3-Clause**
Noto Sans SC, Noto Sans Tamil (subset)	**model**	**SIL OFL 1.1**
Ubuntu font	**model**	**Ubuntu Font Licence 1.0**

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No [x] No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

HARRISH

1/11/2026